

TECHNOLOGY STACK JUSTIFICATION

AlgoNova – AI-Powered DSA & System Design Mastering Platform

AlgoNova follows a decoupled client-server architecture. The selected stack supports an interactive React interface, modular backend APIs, persistent cloud storage, AI-powered features, secure authentication and specialized coding/System Design workflows. The selections below are based on the technologies documented in the project README.

Layer / Tool	Selected Technology	Justification
Frontend Framework	React 19 + Vite	Supports component-based interactive UI development with a fast development and build workflow.
UI Styling	Tailwind CSS	Supports rapid, consistent implementation of the custom Neo-Brutalist design system.
Icons	Lucide React	Provides a consistent icon set integrated with React components.
Animation	Framer Motion	Supports smooth UI and learning-oriented animations.
Code Editor	@monaco-editor/react	Provides a browser-based editor for coding practice and interview workflows.
System Design Canvas	@excalidraw/excalidraw	Provides the interactive whiteboard used to create architecture diagrams.
PDF Processing	pdfjs-dist	Enables client-side resume PDF text extraction.
Backend Runtime	Node.js	Provides the server-side JavaScript runtime for APIs and application logic.
Backend Framework	Express.js	Supports modular routes and middleware for authentication, AI, resume and other APIs.
Database	MongoDB Atlas + Mongoose	Provides cloud document storage and structured database access for application and progress data.
AI Engine	Google Gemini API	Powers interview evaluation, System Design critique, resume analysis/job matching and coding feedback.
Authentication	Google OAuth 2.0 + JWT	Combines convenient Google authentication with protected JWT-based API access.
Password Security	bcryptjs	Hashes passwords for manually registered accounts.
API Security	express-rate-limit + cors + dotenv	Provides rate limiting, cross-origin control and environment-based configuration.
Email	Nodemailer	Supports email workflows such as password reset communication.
State Management	React Context API	Provides shared application state without an additional state-management framework.
Deployment	Vercel + Render	Separates frontend delivery from backend API hosting as documented in the project README.

Overall Technology Justification

The stack forms a coherent full-stack architecture: React and Vite handle the presentation layer; Node.js and Express.js provide backend services; MongoDB Atlas with Mongoose provides persistent data storage; and Google Gemini supplies the core generative-AI capabilities. Monaco Editor, Excalidraw, Framer Motion and pdfjs-dist directly support the platform's coding, System Design, visualization and resume-analysis requirements. The README documents Vercel for the frontend and Render for the backend deployment.